The 4-RGDD of type 10^4 and three mutually orthogonal Latin squares of order 10: an equivalence, what is known, and a computer search

Abstract

In the existence theorems of Wei and Ge for 4-RGDDs [25, Theorem 4.21], [26, Theorem 5.2] the type 10^4 is one of the possible exceptions. We were asked to construct a 4-RGDD of type 10^4 directly, i.e. three mutually orthogonal Latin squares of order 10, or else to prove that none exists. **We achieved neither.** We prove that a 4-RGDD of type 10^4 exists if and only if three mutually orthogonal Latin squares of order 10 exist (Theorem 2.2); the existence of three MOLS(10) is a famous open problem, stated as unresolved in work published in 2026 [3]. A construction would settle it one way and a non-existence proof the other, so neither can be expected from routine methods. What this note does contain: (i) complete proofs of the equivalence and of two obstructions to the usual direct constructions (no 4-GDD of type 10^4 admits a group of order 10 fixing three of its groups and acting regularly on them; in particular none admits an abelian point-regular group, so none is cyclic); (ii) a verified incomplete 4-RGDD of type 10^4 with a hole of type 2^4, obtained from Brouwer's four MOLS of order 10 with a hole of order 2, and a proof that filling holes cannot help: a 4-IRGDD of type $(10, h)^4$ has $h \leq 2$, for $h = 2$ the hole can never be filled, and $h = 1$ is equivalent to the original problem (in particular a hole of size 4, which three MOLS of order 4 could fill, cannot occur: there are not even two incomplete MOLS of order 10 with a common 4×4 hole); (iii) an independent verification of the near triple of Egan and Wanless, which in design language is a 4-GDD of type 10^4 with seven pairwise disjoint parallel classes (a resolution needs ten); (iv) a computer search in which 198,464 Latin squares of order 10 were each tested *exhaustively* for membership in a triple of MOLS; none is.

1 Introduction

The existence theorem for 4-RGDDs, in the form updated by Wei and Ge [25, Theorem 4.21], [26, Theorem 5.2], states that the necessary conditions for a 4-RGDD of type h^u,

$$u \geq 4, \qquad hu \equiv 0 \pmod 4, \qquad h(u-1) \equiv 0 \pmod 3,$$

are sufficient except for $(h, u) \in \{(2, 4), (2, 10), (3, 4), (6, 4)\}$ and a finite list of possible exceptions; for $h = 10$ the possible exceptions are $u \in \{4, 94\}$ [26, Theorem 5.2]. A 4-RGDD of type 10^4 has

$$v = 40 \text{ points}, \qquad b = 100 \text{ blocks}, \qquad r = 10 \text{ parallel classes of 10 blocks each},$$

and it satisfies the necessary conditions ($40 \equiv 0 \pmod 4$, $30 \equiv 0 \pmod 3$).

The task and its outcome. The request was to construct such a design directly — equivalently, three pairwise orthogonal Latin squares of order 10 — and, if that could not be done, to give a proof. **No such design is constructed here, and no proof of non-existence is given.** By

Theorem 2.2 either one would decide whether $N(10) \geq 3$, one of the best-known open problems about Latin squares ("It is unknown if $N(10) \geq 3$, and determining the value of $N(10)$ is one of the most prominent unsolved problems concerning MOLS" [3, Section 1]). We do not present heuristic arguments as proofs. What can be proved, and what we found, is organised as follows: Section 2 proves the equivalence; Section 3 summarises what is known about three MOLS(10) and explains why we cannot supply a proof either way; Section 4 proves two obstructions relevant to direct constructions; Section 5 gives explicit, verified near misses; Section 6 describes our computer search. The script `verify_10_4.py` checks every object printed here from the definitions, and can check any candidate triple of squares.

2 Definitions and the equivalence

Throughout $[g] = \{0, 1, \ldots, g-1\}$. A *$k$-GDD of type g^u* is a triple $(X, \mathcal{G}, \mathcal{B})$ where X is a set of gu points, $\mathcal{G}$ is a partition of X into u *groups* of size g, and $\mathcal{B}$ is a set of k-subsets of X (*blocks*) such that every pair of points from distinct groups lies in exactly one block and no block contains two points of the same group. A *parallel class* is a set of blocks partitioning X; the GDD is *resolvable* (a k-RGDD) if $\mathcal{B}$ can be partitioned into parallel classes. A $\mathrm{TD}(k, g)$ is a k-GDD of type g^k, and an $\mathrm{RTD}(k, g)$ is a resolvable one.

A *Latin square* of order g is a $g \times g$ array with entries in $[g]$ in which every symbol occurs once in each row and once in each column. Latin squares A and B are *orthogonal* ($A \perp B$) if the g^2 ordered pairs $(A(i,j), B(i,j))$ are distinct; MOLS are pairwise orthogonal Latin squares, and $N(g)$ is the largest number of MOLS of order g. A *transversal* of A is a set of g cells containing one cell of each row and each column and each symbol once; a *common transversal* of A and B is a transversal of both. If $A \perp B$ then each symbol class $\{(i,j) : B(i,j) = s\}$ is a transversal of A; conversely every partition of the cells into g transversals of A defines (after labelling the transversals by $[g]$) a Latin square orthogonal to A, an *orthogonal mate* of A.

Lemma 2.1. *In a 4-GDD of type g^4 (that is, a $\mathrm{TD}(4, g)$) every block meets every group in exactly one point and there are exactly g^2 blocks. Every parallel class has g blocks, and a resolution has g parallel classes.*

Proof. A block has four points, no two in one group, and there are four groups. Each block contains $\binom{4}{2} = 6$ pairs from distinct groups; there are $6g^2$ such pairs and each lies in exactly one block. A parallel class covers the $4g$ points by blocks of size 4. □

Theorem 2.2. *Let $g \geq 2$. A 4-RGDD of type g^4 exists if and only if there are three MOLS of order g, if and only if a $\mathrm{TD}(5, g)$ exists. Explicitly: let A, B, C be MOLS of order g, let*

$$X = \{r_i,\ c_i,\ x_i,\ y_i : i \in [g]\}$$

with the four groups $\{r_i\}$, $\{c_i\}$, $\{x_i\}$, $\{y_i\}$, and for every cell (i,j) let

$$\beta_{ij} = \{\, r_i,\ c_j,\ x_{A(i,j)},\ y_{B(i,j)} \,\}.$$

Then the blocks β_{ij} form a 4-GDD of type g^4, and the sets $\mathcal{P}_s = \{\beta_{ij} : C(i,j) = s\}$, $s \in [g]$, form a resolution of it. Every 4-RGDD of type g^4 arises in this way.

Proof. From squares to the design. The pair $\{r_i, c_j\}$ lies only in β_{ij}; the pair $\{r_i, x_s\}$ lies in β_{ij} exactly when $A(i,j) = s$, which happens for exactly one j because row i of A contains s once; the same argument with columns and with B handles $\{c_j, x_s\}$, $\{r_i, y_t\}$, $\{c_j, y_t\}$; and $\{x_s, y_t\}$ lies in β_{ij}

exactly when $(A(i,j), B(i,j)) = (s,t)$, which happens once because $A \perp B$. No block contains two points of a group. The cells with $C(i,j) = s$ form a common transversal of A and B: one cell in each row and column because C is Latin, and each symbol of A (of B) once because $C \perp A$ ($C \perp B$). Hence the blocks of $\mathcal{P}_s$ contain each r_i, c_j, x_s, y_t exactly once, i.e. $\mathcal{P}_s$ is a parallel class, and the $\mathcal{P}_s$ partition the blocks.

From the design to squares. Let $(X, \mathcal{G}, \mathcal{B})$ be a 4-RGDD of type g^4; name the groups and their points $\{r_i\}, \{c_j\}, \{x_s\}, \{y_t\}$ and the parallel classes $\mathcal{P}_0, \ldots, \mathcal{P}_{g-1}$ (Lemma 2.1). By Lemma 2.1 the unique block through r_i and c_j has the form $\{r_i, c_j, x_s, y_t\}$; put $A(i,j) = s$, $B(i,j) = t$ and let $C(i,j)$ be the index of the parallel class containing that block. The pair conditions for $\{r_i, x_s\}$ and $\{c_j, x_s\}$ say that A is Latin, likewise for B, and the condition for $\{x_s, y_t\}$ says $A \perp B$. A parallel class contains each r_i and each c_j once, so C is Latin, and it contains each x_s and each y_t once, so the pairs (A, C) and (B, C) take every value once: $C \perp A$ and $C \perp B$.

TD$(5, g)$. Given a resolution $\mathcal{P}_0, \ldots, \mathcal{P}_{g-1}$, adjoin a fifth group $\{\infty_0, \ldots, \infty_{g-1}\}$ and add ∞_s to every block of $\mathcal{P}_s$; since $\mathcal{P}_s$ partitions X, every pair $\{z, \infty_s\}$ lies in exactly one block, so this is a TD$(5, g)$. Conversely, in a TD$(5, g)$ the blocks through a point ∞_s of the fifth group, with ∞_s removed, form a parallel class of the TD$(4, g)$ obtained by deleting that group, and these classes partition its blocks. □

Corollary 2.3. *For $g \geq 2$ a 4-RGDD of type g^4 exists if and only if $N(g) \geq 3$. As $N(2) = 1$, $N(3) = 2$, $N(6) = 1$ (Tarry [24]) and $N(g) \geq 3$ for every other $g \neq 10$ ([1]; consistently, [26, Theorem 5.2] lists $g = 10$ as the only possible exception of type g^4), such a design exists for all $g \notin \{2, 3, 6, 10\}$ and does not exist for $g \in \{2, 3, 6\}$ — exactly the pattern of exceptions $(2,4), (3,4), (6,4)$ and possible exception $(10,4)$ in [25, 26]. For $g = 10$ the question is equivalent to $N(10) \geq 3$.*

Corollary 2.4. *Every 4-GDD of type 10^4 is the design $\{\beta_{ij}\}$ of a pair of orthogonal Latin squares (A, B) of order 10 (such pairs exist [21, 5]), and its parallel classes are exactly the sets $\{\beta_{ij} : (i,j) \in T\}$ with T a common transversal of A and B. Hence a 4-GDD of type 10^4 is resolvable if and only if the 100 cells can be partitioned into 10 common transversals of A and B.*

Proof. The first statement is the second part of the proof of Theorem 2.2, which does not use the resolution to define A and B. The blocks over a set T of cells contain every point exactly once if and only if T is a common transversal. □

So 4-GDDs of type 10^4 exist in abundance; the entire difficulty is resolvability.

3 What is known, and why we cannot give a proof

Bounds. $N(10) \geq 2$: Parker [21] found two orthogonal Latin squares of order 10 (see also [4, 5]). $N(10) \leq 6$: there is no projective plane of order 10 [17] (see also the independent searches [23, 2]), so $N(10) < 9$, and together with a completion result for nets due to Bruck [9] this gives $N(10) \leq 6$, the best known upper bound [3, Section 2.4], [11].

Partial non-existence results. Each of the following excludes triples of MOLS(10) with a special structure (statements as summarised in [3]).

- Mann [18]: a Latin square of order 10 with a 5×5 Latin subsquare has no orthogonal mate (Proposition 4.1 below).
- Parker [22]: two orthogonal Latin squares of order 10 with orthogonal 3×3 Latin subsquares cannot be part of an orthogonal triple.

- McKay, Meynert and Myrvold [19]: every Latin square in a triple of MOLS(10) has trivial symmetry group.
- Myrvold [20], Bright, Keita and Stevens [3]: if one square of a triple has a 4×4 Latin subsquare, the other two fall into one of 28 "mate pattern type pairs"; Myrvold excluded 20 of them, and [3] re-verified this with a SAT solver and found orthogonal pairs in all eight remaining cases, so those cases cannot be excluded by looking at pairs alone.
- Gill and Wanless [13]: the rank of the code generated by any pair of squares of a triple of MOLS(10) is exactly 37; see also Delisle [10] and Dukes and Howard [11].
- Zaikin, Vatutin and Bright [28]: no extended self-orthogonal diagonal Latin square of order 10 lies in an orthogonal triple.

Near misses. Brouwer [8]: four MOLS of order 10 with a common hole of order 2, i.e. a $\mathrm{TD}(6,10) - \mathrm{TD}(6,2)$ (Section 5.1). Egan and Wanless [12]: squares L_1, L_2, L_3 of order 10 with $L_1 \perp L_2$, $L_1 \perp L_3$ and 91 of the 100 ordered pairs distinct when L_2 and L_3 are superimposed; hence L_1 and L_2 have seven pairwise disjoint common transversals, improving the four of Brown, Hedayat and Parker [7] (Section 5.2). Zaikin and Kochemazov [27]: three diagonal Latin squares of order 10 satisfying the orthogonality condition on a common set of 73 cells.

Why we cannot give a proof.

1. By Theorem 2.2, a proof that no 4-RGDD of type 10^4 exists is a proof that $N(10) = 2$, and a construction proves $N(10) \geq 3$. Either would settle the open problem above. We have neither.
2. All the standard necessary conditions are satisfied: the divisibility conditions hold, the underlying 4-GDD of type 10^4 exists (Corollary 2.4), and so do incomplete versions with a small hole (Section 5.1). Non-existence therefore cannot follow from these conditions.
3. The proof that $N(10) < 9$ [17] relies on the binary code of a projective plane and still needed a very large computation. The methods behind the partial results above (symmetry, subsquares, codes of pairs) each exclude only special configurations.
4. A naive exhaustive search would test one square from each of the 34,817,397,894,749,939 main classes of Latin squares of order 10 [19]. (Membership in a triple is a main-class invariant: a square is a component of a triple if and only if its $\mathrm{TD}(3,10)$ extends to a $\mathrm{TD}(5,10)$, and this does not depend on which of its three groups are called rows, columns and symbols.) At the speed of our own complete test (about 27 squares per second per core, Section 6) this would take about 4.1×10^7 core-years.

4 Two obstructions with complete proofs

Proposition 4.1 (Mann [18])**.** *A Latin square of order $4t+2$ containing a Latin subsquare of order $2t+1$ has no transversal. In particular no Latin square of order 10 with a 5×5 Latin subsquare has an orthogonal mate, so none can be one of the squares A, B, C of Theorem 2.2.*

Proof. Let the subsquare occupy rows R and columns K with symbol set S, $|R| = |K| = |S| = m = 2t+1$, and let bars denote complements ($|\bar{R}| = |\bar{K}| = |\bar{S}| = m$). Rows in R carry the symbols

of S in the columns K, hence the symbols of $\bar{S}$ in the columns $\bar{K}$; likewise columns in K carry $\bar{S}$ in the rows $\bar{R}$; consequently the cells $\bar{R} \times \bar{K}$ carry only symbols of S. Let T be a transversal and let a, b, d be the numbers of its cells in $R \times K$, $R \times \bar{K}$, $\bar{R} \times \bar{K}$. Counting the cells of T in the rows R, in the columns $\bar{K}$, and those with a symbol in S gives

$$a + b = m, \qquad b + d = m, \qquad a + d = m.$$

The first and third give $b = d$, and then $2b = m$, which is impossible since m is odd. □

A *complete mapping* of a group H is a bijection $\theta\colon H \to H$ such that $x \mapsto x\theta(x)$ is also a bijection; equivalently the cells $(x, \theta(x))$ form a transversal of the Cayley table $(x, y) \mapsto xy$.

Proposition 4.2. *No group of order* 10 *has a complete mapping.*

Proof. A group H of order 10 has a normal subgroup N of order 5 (its Sylow 5-subgroup); let $\varepsilon\colon H \to H/N \cong \mathbb{Z}_2$, written additively. If θ were a complete mapping, then summing $\varepsilon(x\theta(x)) = \varepsilon(x) + \varepsilon(\theta(x))$ over $x \in H$ would give $\sum_{y \in H} \varepsilon(y) = 2\sum_{y \in H} \varepsilon(y)$ in $\mathbb{Z}_2$, because $x \mapsto x\theta(x)$ and θ are bijections. But $\sum_{y \in H} \varepsilon(y) = |H \setminus N| = 5 \equiv 1$, a contradiction. (This is a special case of the Hall–Paige theorem [14].) □

Lemma 4.3. *Let $(X, \{G_1, G_2, G_3\}, \mathcal{B})$ be a* $\mathrm{TD}(3, m)$ *and let H be a group of order m acting on it by automorphisms that map each G_i onto itself and act regularly on each G_i. Then H has a complete mapping.*

Proof. Fix $p_i \in G_i$ and write (h, i) for the point $h(p_i)$; so $g \in H$ maps (h, i) to (gh, i). For $a \in H$ let $\{(1,1), (a,2), (\theta(a), 3)\}$ be the block through $(1,1)$ and $(a,2)$. The map θ is injective, because the pair $\{(1,1), (\theta(a), 3)\}$ lies in only one block, hence bijective. Every block contains exactly one point $(h, 1)$, and h^{-1} maps it to a block through $(1,1)$; so the blocks are exactly $\{(h,1), (ha, 2), (h\theta(a), 3)\}$ for $h, a \in H$. The pair $\{(y,2), (z,3)\}$ lies in the block with parameters (h, a) if and only if $h = ya^{-1}$ and $a^{-1}\theta(a) = y^{-1}z$. Since every such pair lies in exactly one block, $a \mapsto a^{-1}\theta(a)$ is a bijection. With $\sigma(x) = \theta(x^{-1})$ we get $x\sigma(x) = a^{-1}\theta(a)$ for $a = x^{-1}$, so σ is a complete mapping. □

Corollary 4.4. *(a) No* 4*-GDD of type* 10^4*, resolvable or not, admits a group H of order* 10 *of automorphisms that maps three of its groups onto themselves and acts regularly on each of them. In particular no* $\mathrm{TD}(k, 10)$ *with $k \geq 3$, and so no* 4*-RGDD of type* 10^4*, comes from a* $(10, k; 1)$*-difference matrix over a group of order* 10 *(compare [25, Lemma 2.8], which gives such transversal designs with $k = 2$ only).*

(b) No 4*-GDD of type* 10^4 *admits an abelian group of automorphisms acting regularly on its* 40 *points. In particular none is cyclic, and none is obtained by developing base blocks under a regular action of one of the abelian groups of order* 40*, namely* $\mathbb{Z}_{40}$*,* $\mathbb{Z}_{20} \times \mathbb{Z}_2 \cong \mathbb{Z}_{10} \times \mathbb{Z}_4$ *and* $\mathbb{Z}_{10} \times \mathbb{Z}_2^2$*.*

(c) If G is any group of automorphisms of a 4*-GDD of type* 10^4 *acting regularly on the* 40 *points, then the kernel K of the action of G on the four groups has order* 5 *and G/K is dihedral of order* 8*, acting on the four groups as the symmetry group of a square.*

Proof. An automorphism maps groups onto groups, since two points lie in a common group if and only if no block contains both. (a) Deleting a fourth group gives a $\mathrm{TD}(3, 10)$ to which Lemma 4.3 applies, contradicting Proposition 4.2. A $(10, k; 1)$-difference matrix over H yields a $\mathrm{TD}(k, 10)$ on which H acts by translation, fixing every group and regularly on each. (b), (c) Let G act regularly;

then it acts transitively on the four groups. Let Γ be the group containing a point p_0 and H its stabiliser, so $|H| = 10$, H acts regularly on Γ, and the kernel K of the action on the groups is the largest normal subgroup of G contained in H. Now G/K embeds in S_4 and is transitive on four objects, so $|G/K|$ is a multiple of 4 dividing $\gcd(40, 24) = 8$. If $|G/K| = 4$ then $K = H$ fixes every group and acts on each freely, hence regularly, which (a) forbids. So $|K| = 5$ and $|G/K| = 8$; the only subgroups of order 8 of S_4 are its Sylow 2-subgroups, dihedral of order 8. If G is abelian then H is normal, so $K = H$, which is impossible; this proves (b). □

Remark 4.5. Combined with the computer-assisted theorem of [19] quoted in Section 3 (every square of a triple of MOLS(10) has trivial symmetry group), much more follows. Let $\varphi \neq 1$ be an automorphism of the 4-GDD underlying a 4-RGDD of type 10^4 (the resolution need not be preserved) that maps each of the four groups onto itself. Then φ permutes the rows, the columns and the symbols of the square A of Theorem 2.2 and maps the block of cell (i, j) to the block of the image cell, so it induces an autotopism of A; this autotopism is non-trivial, since otherwise φ would fix every block through every pair r_i, c_j and hence every point. As A is a component of a triple, this contradicts [19]. So *every non-identity automorphism of a* 4*-RGDD of type* 10^4 *moves at least one of its groups*; in particular the subgroup K in (c) cannot exist, and *no* 4*-RGDD of type* 10^4 *admits a point-regular automorphism group.* These two statements rest on the computation of [19], which we have not repeated.

5 Explicit near misses (verified)

5.1 An incomplete 4-RGDD of type 10^4 with a hole of type 2^4

Brouwer [8] gave four MOLS of order 10 with a common hole of order 2 (Figure 1, transcribed from [8, Fig. 1]; entry $ijkl$ in cell (r, c) means that the four squares have symbols i, j, k, l there; rows and columns are numbered $0, \ldots, 9$, the hole is $\{8, 9\} \times \{8, 9\}$, and the hole symbols are $8, 9$).

5039	8225	4568	9753	6812	0390	2487	7976	3141	1604
9460	6109	8336	5678	4947	7823	1091	3584	0252	2715
6748	9571	7219	8007	0685	5954	4830	2192	1363	3426
8114	7458	9642	4329	3293	1786	6965	5801	2070	0537
2856	4794	6083	3932	1479	8661	0128	9317	7505	5240
0903	3867	5495	7180	9024	2549	8772	1238	4616	6351
4281	1910	0874	6596	2308	9135	3659	8443	5727	7062
7697	5382	2921	1845	8550	3018	9206	0769	6434	4173
1522	2633	3700	0411	5166	6277	7344	4055	–	–
3375	0046	1157	2264	7731	4402	5513	6620	–	–

Figure 1: Four MOLS of order 10 with a hole of order 2 (Brouwer [8]).

Proposition 5.1. *Let* A, B, C *be the first three of the squares in Figure 1, defined on the* 96 *cells outside the hole. With* X *and* β_{ij} *as in Theorem 2.2 and*

$$Y = \{r_8, r_9, c_8, c_9, x_8, x_9, y_8, y_9\},$$

the 96 *blocks* β_{ij} *have the following properties.*

(i) Every pair of points from distinct groups that are not both in Y lies in exactly one block, no pair inside Y lies in a block, and every block meets Y in at most one point. (This is a 4-IGDD of type $(10,2)^4$ in the notation of [25].)

(ii) For $s \in \{0,\dots,7\}$ the blocks of the cells with $C(i,j) = s$ form a parallel class (10 blocks); for $s \in \{8,9\}$ they form a holey parallel class (8 blocks partitioning $X \setminus Y$). So the design is a 4-IRGDD of type $(10,2)^4$.

(iii) The hole cannot be filled: there is no set of blocks of size 4 which, added to these 96 blocks, yields a 4-GDD of type 10^4, let alone a 4-RGDD.

Proof. (i) and (ii) say that A, B, C are incomplete MOLS with a common hole (every row and column outside the hole contains every symbol once, the rows and columns of the hole contain each of $0,\dots,7$ once outside the hole, and each pair of squares shows every ordered pair of symbols not both in $\{8,9\}$ exactly once); the argument of Theorem 2.2 translates this into (i) and (ii). All of this is checked by `verify_10_4.py`, which in fact verifies that all four squares form a $\mathrm{TD}(6,10) - \mathrm{TD}(6,2)$. (iii) The pairs not yet covered are exactly the 24 pairs of points of Y in distinct groups. An added block may contain no pair that is already covered, so all its points lie in Y; hence the added blocks would form a 4-GDD of type 2^4 on Y. By the second part of the proof of Theorem 2.2 (with $g = 2$) such a design gives two orthogonal Latin squares of order 2, which do not exist (see also [6]). □

Thus a resolvable design on 40 points that agrees with a 4-RGDD of type 10^4 on all but the 4 blocks inside an 8-point hole exists, and the standard "filling in holes" construction fails here only because $N(2) = 1$. No other incomplete design can do better:

Proposition 5.2. *Let $1 \le h \le 9$ and suppose that a 4-IRGDD of type $(10,h)^4$ exists, with hole Y ($|Y| = 4h$). Then:*

(i) $h \le 2$;

(ii) if $h = 2$, no set of blocks can be added to obtain a 4-GDD of type 10^4;

(iii) if $h = 1$, adding Y as a block gives a 4-RGDD of type 10^4.

So the "filling in holes" approach can only produce a 4-RGDD of type 10^4 from a design that is already one with a block removed.

Proof. A point of Y forms a pair with each of the $3(10-h)$ points outside Y and outside its own group, and every block through it contains three of these (a block meets Y at most once); so it lies in $10 - h \ge 1$ blocks, all of them in (full) parallel classes, since the holey ones avoid Y. Likewise a point outside Y lies in 10 blocks, one in each parallel class and one in each holey parallel class; hence there are $10 - h$ parallel and h holey parallel classes. A parallel class has 10 blocks, each meeting Y in at most one point, and it covers the $4h$ points of Y; so $4h \le 10$, which is (i). (ii) is the argument of Proposition 5.1(iii), which used only the hole. (iii) If $h = 1$ the uncovered pairs are the six pairs inside the 4-set Y, which meets every group once, so Y can be added as a block; added to the unique holey parallel class, it completes a parallel class. □

Remark 5.3 (Holes of size $h \ge 3$, in particular $h = 4$). For $h \in \{4,5,7,8,9\}$ the design needed to fill the hole does exist: $N(h) \ge 3$, so by Theorem 2.2 there is a 4-RGDD of type h^4; for $h = 4$ it comes from three MOLS of order 4. The obstruction is therefore the incomplete design, not the filling. For $h = 4$ the hole Y has 16 points, and every point of Y lies in $10 - h = 6$ blocks, all in

full parallel classes; but a full parallel class has only 10 blocks, each with at most one point of Y, so it cannot cover the 16 points of Y.

Even the non-resolvable version fails: there are not even two incomplete MOLS of order 10 with a common 4×4 hole (a $\mathrm{TD}(4,10) - \mathrm{TD}(4,4)$). Let L, M be such squares, with hole rows R, hole columns K and hole symbols S, $|R| = |K| = |S| = 4$. A hole symbol never occurs in a hole row or a hole column, so in each of the six rows outside R the four hole symbols of L lie in the six columns outside K: L has a hole symbol in 24 of the 36 cells of $\bar{R} \times \bar{K}$, and so has M. Hence at least $24 + 24 - 36 = 12$ cells carry a pair of hole symbols, which an incomplete pair must not contain. The same count gives the standard bound: k incomplete MOLS of order n with a common hole of size h ($0 < h < n$) need $k\,h(n-h) \le (n-h)^2$, i.e. $n \ge (k+1)h$. For $n = 10$ this means $h \le 3$ for a pair and $h \le 2$ for a triple.

The bound also disposes of the variants with several filled sub-designs, as in [25, Construction 4.3] with $10 = h_1 + h_2 + \cdots$: removing a filled sub-design of type h_i^4, whose parallel classes extend to parallel classes of the whole design, leaves a 4-IRGDD of type $(10, h_i)^4$. So every $h_i \le 2$, a part $h_i = 2$ cannot be filled, and only $h_i = 1$ (single blocks) remains.

As an independent check, a SAT solver (CaDiCaL via PySAT, script `imols_sat.py`) reports within a second that no two and no three incomplete MOLS of order 10 with a common 4×4 hole exist, and no three with a common 3×3 hole; on the other hand it finds at once an incomplete Latin square of order 10 with a 4×4 hole, two incomplete MOLS of order 12 with a common 4×4 hole, and two incomplete MOLS of order 9 with a common 3×3 hole (all solutions checked by the script).

5.2 The near triple of Egan and Wanless

Egan and Wanless [12, Section 8] examined the Latin squares of order 10 with an autoparatopism group of order at least 3 (catalogued in [19]) and found the squares A, B, C of Figure 2, which are closer to a triple of MOLS than any previously published example.

0 8 9 7 5 6 4 2 3 1	0 7 8 9 1 2 3 4 5 6	0 7 8 9 1 2 3 4 5 6
9 1 4 6 2 7 3 8 0 5	9 0 6 1 8 3 2 5 4 7	6 4 2 8 9 5 1 3 7 0
7 4 2 5 1 3 8 6 9 0	7 2 0 4 3 9 1 8 6 5	4 9 5 3 2 7 6 0 1 8
8 6 5 3 9 2 1 0 4 7	8 5 3 0 2 1 7 6 9 4	5 1 7 6 4 3 8 9 0 2
6 2 1 8 4 0 9 5 7 3	6 9 5 3 0 7 4 2 1 8	3 2 9 0 7 1 5 6 8 4
4 9 3 2 7 5 0 1 6 8	4 1 7 6 5 0 8 9 3 2	1 0 3 7 6 8 2 5 4 9
5 3 7 1 0 8 6 9 2 4	5 4 2 8 9 6 0 3 7 1	2 8 0 1 3 4 9 7 6 5
3 5 0 9 8 4 2 7 1 6	3 6 1 7 4 8 5 0 2 9	9 5 4 2 8 6 0 1 3 7
1 7 6 0 3 9 5 4 8 2	1 8 4 2 6 5 9 7 0 3	7 3 6 5 0 9 4 8 2 1
2 0 8 4 6 1 7 3 5 9	2 3 9 5 7 4 6 1 8 0	8 6 1 4 5 0 7 2 9 3

Figure 2: The squares A (left), B (middle), C (right) of Egan and Wanless [12, Section 8].

Proposition 5.4. *For the squares of Figure 2:*

(i) $A \perp B$, $A \perp C$, and superimposing B and C gives 91 distinct ordered pairs; every repeated pair has C-symbol 7, 8 or 9 [12].

(ii) Consequently the 4-GDD of type 10^4 of the pair (A, B) (Theorem 2.2) contains the seven pairwise disjoint parallel classes $\mathcal{P}_s = \{\beta_{ij} : C(i,j) = s\}$, $s = 0, \ldots, 6$; the remaining 30 blocks contain every point exactly three times.

(iii) *A and B have exactly* 14 *common transversals, and at most* 7 *of them are pairwise disjoint. So this* 4*-GDD has* 14 *parallel classes, no eight of them are pairwise disjoint, and the* 30 *remaining blocks of (ii) cannot be split into three parallel classes: the design is three parallel classes short of a* 4*-RGDD of type* 10^4*.*

(iv) *A has exactly* 1080 *transversals and* 305 *orthogonal mates, and for none of them, M say, is there a Latin square orthogonal to both A and M; so A is not a component of a triple, as already known from [19, 12]; the same holds for B (*932 *transversals,* 5 *mates) and C (*864 *transversals,* 4 *mates). Over all pairs of mates of A,* 91 *is the largest number of distinct ordered pairs.*

Proof. (i)–(iii) are checked by `verify_10_4.py` from the definitions (for (iii) it enumerates all common transversals and finds a largest set of pairwise disjoint ones). For (ii): $C \perp A$, and the classes of the symbols $0, \ldots, 6$ of C contain no repeated pair, so they are common transversals of A and B; each point lies in 10 blocks, seven of them in these classes. (iv) is the complete test of Section 6. □

In the language of designs, this is the best known approximation to a 4-RGDD of type 10^4 that we are aware of: a 4-GDD of type 10^4 with seven pairwise disjoint parallel classes (Brown, Hedayat and Parker [7] had four).

5.3 A pair of orthogonal Latin squares from our search

The best pair found by our search (Section 6) is shown in Figure 3: two orthogonal Latin squares A, B of order 10 and, in the third array, 3 pairwise disjoint common transversals of A and B (cells marked $0, \ldots, 2$).

7 2 1 4 5 8 6 9 0 3	5 6 2 4 1 7 3 8 9 0	· · · · 0 1 · · 2 ·
3 4 5 0 8 6 9 7 1 2	7 5 9 1 6 2 0 4 8 3	· · · 1 · 2 · · · 0
6 5 8 3 9 2 0 1 7 4	4 7 0 2 9 1 5 3 6 8	· · 2 · · · · · 0 1
1 3 4 7 2 9 8 0 6 5	9 8 7 3 4 5 2 0 1 6	1 0 · · · · · · · 2
9 1 3 8 0 7 5 2 4 6	2 1 6 8 3 9 4 7 0 5	0 · · · · · 1 2 · ·
2 0 9 6 7 3 1 4 5 8	0 2 3 7 8 4 6 1 5 9	· · 1 0 2 · · · · ·
8 9 2 5 1 0 4 6 3 7	1 4 5 0 7 8 9 6 3 2	· 2 · · · · 0 1 · ·
5 8 0 1 6 4 7 3 2 9	8 3 4 5 0 6 7 9 2 1	· · 0 2 · · · · 1 ·
0 7 6 2 4 5 3 8 9 1	6 0 8 9 2 3 1 5 7 4	· 1 · · · · 2 0 · ·
4 6 7 9 3 1 2 5 8 0	3 9 1 6 5 0 8 2 4 7	2 · · · 1 0 · · · ·

Figure 3: Orthogonal Latin squares A (left) and B (middle) of order 10, and 3 pairwise disjoint common transversals of A and B (right).

Proposition 5.5. *For the squares of Figure 3, the* 100 *blocks β_{ij} of Theorem 2.2 form a* 4*-GDD of type* 10^4 *with 3 pairwise disjoint parallel classes (the cells marked* $0, \ldots, 2$*). The squares A and B have exactly 3 common transversals, so this* 4*-GDD has exactly 3 parallel classes and is not resolvable; 7 of the ten pairwise disjoint parallel classes of a resolution are missing.*

Proof. `verify_10_4.py` checks that $A \perp B$, checks the 4-GDD and the 3 parallel classes from the definitions, and enumerates all common transversals of A and B; Corollary 2.4 identifies these with the parallel classes. □

6 The computer search

Lemma 6.1. *A Latin square L of order n is a component of a set of three MOLS if and only if the cells admit two partitions $\mathcal{P}$, $\mathcal{Q}$ into transversals of L with $|T \cap U| = 1$ for all $T \in \mathcal{P}$, $U \in \mathcal{Q}$. Equivalently, for some orthogonal mate M of L (given by $\mathcal{P}$) the common transversals of L and M, namely the transversals U of L with $|U \cap T| = 1$ for all $T \in \mathcal{P}$, contain a partition of the cells.*

Proof. If L, M, M' are MOLS take for $\mathcal{P}, \mathcal{Q}$ the symbol classes of M, M'; the condition $|T \cap U| = 1$ says that (M, M') takes every pair of values once. Conversely label the members of $\mathcal{P}$ and $\mathcal{Q}$ by $[n]$ to obtain M and M'. □

Algorithm. For a square L of order 10 the program `mols3_search.c`

1. enumerates all transversals of L by backtracking;
2. enumerates *all* orthogonal mates, i.e. all exact covers of the 100 cells by transversals (Algorithm X [16]: branch on an uncovered cell with the fewest candidates, prune as soon as some uncovered cell has none);
3. for every mate $\mathcal{P}$ computes the set $C(\mathcal{P})$ of common transversals as an intersection of precomputed bit sets $\{U : |U \cap T| = 1\}$, $T \in \mathcal{P}$, and decides by a second exact-cover search whether $C(\mathcal{P})$ contains a partition of the cells;
4. records $\nu(\mathcal{P})$, the largest number of pairwise disjoint members of $C(\mathcal{P})$ (a maximum clique computation), and, for every two mates M, M' of L, the number of distinct ordered pairs $(M(i,j), M'(i,j))$.

Steps 1–3 decide completely whether L is a component of a triple (Lemma 6.1). By Corollary 2.4, $\nu(\mathcal{P})$ is the largest number of pairwise disjoint parallel classes of the 4-GDD of type 10^4 defined by the pair (L, M); $\nu = 10$ would be a 4-RGDD of type 10^4. The pair count in step 4 is the measure used by Egan and Wanless (100 would be a triple, their record is 91).

Validation. The program reproduces the transversal counts of the Cayley tables of $\mathbb{Z}_7$ (133), $\mathbb{Z}_2^3$ (384), $\mathbb{Z}_3^2$ (2241), $\mathbb{Z}_9$ (2025), and finds none for $\mathbb{Z}_8$, $\mathbb{Z}_{10}$ and the dihedral group D_5 (in accordance with Proposition 4.2). It finds triples of MOLS containing the Cayley tables of $\mathbb{Z}_7$, $\mathbb{Z}_2^3$, $\mathbb{Z}_3^2$ and $\mathbb{Z}_9$ (the triples were checked independently). The Cayley tables of $\mathbb{Z}_9$ and $\mathbb{Z}_3^2$ have 2,049,219 and 12,445,836 orthogonal mates respectively (here and below a mate is counted as a partition of the cells into transversals, i.e. up to renaming its symbols), exactly the numbers reported by Egan and Wanless [12, Section 5]. An independent Python implementation (dictionary-based Algorithm X) reproduced, for $\mathbb{Z}_7$ and for six random squares of order 10, the numbers of transversals and mates, the maximum of $|C(\mathcal{P})|$ and the number of mates that extend to a triple (133, 635, 28, 5 for $\mathbb{Z}_7$). Applied to the squares of Egan and Wanless (Section 5.2), the C program reproduces the values 91 and 7 reported in [12].

Searches.

1. *Random squares.* Squares sampled with the Jacobson–Matthews Markov chain [15] (at least 1500 steps between samples).
2. *Prolongations.* For every transversal T of the Cayley tables of $\mathbb{Z}_9$ and $\mathbb{Z}_3^2$, the prolongation to order 10 (move the symbols on T to a new row and column and put the new symbol on T); these squares have many more transversals (1376–1880) and up to 2,592 orthogonal mates.

3. A simulated annealing on the number of transversals was also tried; it did not get beyond about 980 transversals (random squares have about 800) and was abandoned.

family	squares	squares with a mate	mates	mates with $C(\mathcal{P}) \neq \emptyset$	$\max \lvert C(\mathcal{P}) \rvert$	$\max \nu$	best pairs	triples
random	194,198	117,730	198,229	22,557	3	3	78	0
prolongations, $\mathbb{Z}_9$	2,025	1,971	134,136	32,076	1	1	74	0
prolongations, $\mathbb{Z}_3^2$	2,241	1,368	264,384	116,640	2	1	75	0

Table 1: Exhaustive per-square tests. "mates": orthogonal mates found (all of them, for every square); "$C(\mathcal{P}) \neq \emptyset$": mates M for which L and M have at least one common transversal; ν: largest number of pairwise disjoint common transversals (disjoint parallel classes of the 4-GDD of type 10^4 of (L, M); 10 would be a 4-RGDD); "best pairs": largest number of distinct ordered pairs $(M(i,j), M'(i,j))$ over two mates $M \neq M'$ of one square (100 would be a triple).

Interpretation. Typical squares of order 10 have mates (in our random sample 60.6% have one and the average number of mates is 1.02, in line with the estimates of [19] quoted in [12, Section 5]: about 60%, and an expectation above 1), but the pairs they form have almost no common transversals: a 4-GDD of type 10^4 obtained from such a pair has at most three disjoint parallel classes in our search, far from the ten needed. This agrees with [3, Table 3], where none of several hundred orthogonal pairs found by a SAT solver had more than two common transversals, and is much weaker than the best known near miss (seven disjoint parallel classes and 91 distinct pairs for the squares of Egan and Wanless, Section 5.2). These are statistics, not evidence of non-existence: the squares tested are a vanishing fraction of the 3.48×10^{16} main classes.

7 Conclusion

A 4-RGDD of type 10^4 is the same object as three MOLS of order 10 (Theorem 2.2). We could not construct one, and no proof of non-existence is known; the case $(h, u) = (10, 4)$ in [25, 26] is precisely the open problem "$N(10) \geq 3$?". We proved that such a design cannot be obtained from a difference matrix over a group of order 10 or by developing base blocks under an abelian point-regular group (Corollary 4.4), that a point-regular group would have to be an extension of $\mathbb{Z}_5$ by the dihedral group of order 8 (and, by [19], cannot exist at all), and that no incomplete design of this kind (such as Brouwer's, with a hole of type 2^4) can be completed except in the trivial case of a hole of size one (Proposition 5.2); in particular a hole of size 4, which three MOLS of order 4 could fill, cannot occur at all (Remark 5.3). The closest known approximation, from the squares of Egan and Wanless, is a 4-GDD of type 10^4 with seven pairwise disjoint parallel classes; our own search, which tested 198,464 squares exhaustively, got no further than three disjoint parallel classes (Proposition 5.5). The explicit near misses above are verified by `verify_10_4.py`; a candidate triple found by any other means can be checked with `python3 verify_10_4.py triple FILE`, which builds the 4-RGDD of Theorem 2.2 and checks it from the definition.

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